

Homework 3

Reading

Please read Section 1.1 of our textbook, stopping after Example 1.2 (and before Section 1.1.1).

Preparing to finish Quiz 1

In question 2 of Quiz 1, you were thinking about a new kind of license plate. The license plates have three letters and three numbers. You can put the letters and numbers in any order. For example, you could have license plates A1B2C3, 314PIE, or EEE271.

I saw groups trying two strategies for counting the number of possible license plates. Both strategies used the multiplication rule:

- Groups using the first strategy first choose the three slots where the letters go. Then they put letters in order (left to right) in those three slots and numbers in order (left to right) in the other three slots.
- Groups using the second strategy first choose three letters and three numbers (each without an order) and then choose an order to write these six things in.

Prepare to finish question 2 of the quiz by answering the following two questions. The first question will help you think about the first strategy and the second question will help you think about the second strategy.

1. How many ways are there to choose three slots out of the six slots on the license plate? What are Durrett's name and notation for this kind of counting?

There are $C_{6,3} = 20$ ways to choose three slots from the set of six slots on the license plate. Durrett calls this kind of counting "combinations."

2. How many ways are there to choose a set of three distinct letters (unordered) out of the set of 26 letters?

There are $C_{26,3} = 2,600$ ways to choose a set of three distinct letters out of the set of 26 letters.

Probability and counting

3. A poker hand is five cards from a standard deck of 52 cards. In our second class, you all figured out that there are $C_{52,5} = 2,598,960$ possible poker hands. If you are dealt a poker hand from a shuffled deck, all of these hands are equally likely. The sample space Ω of (equally likely) outcomes for the chance experiment of being dealt a random poker hand is the set of these $C_{52,5}$ hands. (Don't try to list all the elements of Ω !)

- a. How many poker hands do not have an ace? (There are four aces in a standard deck.) Use Durrett's standard notation for this number and calculate its exact value.

To get a poker hand without an ace, we choose our 5 cards from the 48 cards that are not aces. There are $C_{48,5} = 1,712,304$ ways to make this choice.

- b. How many poker hands have at least one ace?

The remaining $C_{52,5} - C_{48,5} = 2,598,960 - 1,712,304 = 886,656$ poker hands have at least one ace.

- c. What is the probability of the event "you are dealt a poker hand with at least one ace"?

The probability is $886,656 / 2,598,960 \approx 0.341$. The probability of having at least one ace in your hand is a bit more than 1 in 3. That probability may sound surprisingly high, but all our steps are correct!

4. A long time ago, every household had a book called a "phone book." The book listed the telephone numbers of everyone living in their area. For example, a household in Frederick would have a Frederick County telephone book with a list of every house's telephone number. (In those days, people did not have individual telephone numbers. There was one telephone for the whole house, connected to the telephone network by a wire.)

- a. Imagine the chance experiment of opening the telephone book to a random page, picking a random entry on that page, and looking at the last digit of the telephone number in that entry. What is a reasonable sample space Ω for this experiment? To keep with our first examples of probability, the outcomes in your sample space should be equally likely.

A reasonable sample space is $\Omega = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

- b. What is the probability of the event "the last digit is a multiple of 3"?

The event A consists of the following outcomes: $A = \{0, 3, 6, 9\}$. (We should consider 0 to be a multiple of 3, since $0 = 3(0)$.) The probability $P(A)$ of this event is $4/10 = 0.4$, since the outcomes in Ω are equally likely. That will not always be the case!

- c. What is the probability of the event "the last digit is greater than 7"?

The event B consists of the following outcomes: $B = \{8, 9\}$. The probability $P(B)$ of this event is $2/10 = 0.2$.

5. Suppose we roll a red six-sided die and a green six-sided die. We think of these rolls together as our chance experiment. Durrett describes a sample space Ω (with all outcomes equally likely, assuming fair dice) in Example 1.2.

- a. List the outcomes in the event "the number on the red die is larger than the number on the green die."

The outcomes in this event A are the following: $A = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3), (5, 1), (5, 2), (5, 3), (5, 4), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5)\}$.

- b. What is the probability of this event?

There are $1 + 2 + 3 + 4 + 5 = 15$ outcomes in A. Since the outcomes in the sample space Ω are equally likely, the probability $P(A)$ is $15/36 = 5/12 \approx 0.417$.

6. We again imagine the chance experiment of rolling two dice as in question 5 and Example 1.2.

a. List the outcomes in the sample space where the two numbers differ by at most 1.

The outcomes in this event B are the following: $B = \{(1, 1), (1, 2), (2, 1), (2, 2), (2, 3), (3, 2), (3, 3), (3, 4), (4, 3), (4, 4), (4, 5), (5, 4), (5, 5), (5, 6), (6, 5), (6, 6)\}$.

b. What is the probability that the two numbers you roll differ by at most 1?

There are $2 + 3 + 3 + 3 + 3 + 2 = 16$ outcomes in B. Since the outcomes in the sample space Ω are equally likely, the probability $P(B)$ is $16/36 = 4/9 \approx 0.444$.

It is a bit more likely that your two rolls differ by at most one than that the red roll is higher than the green roll.